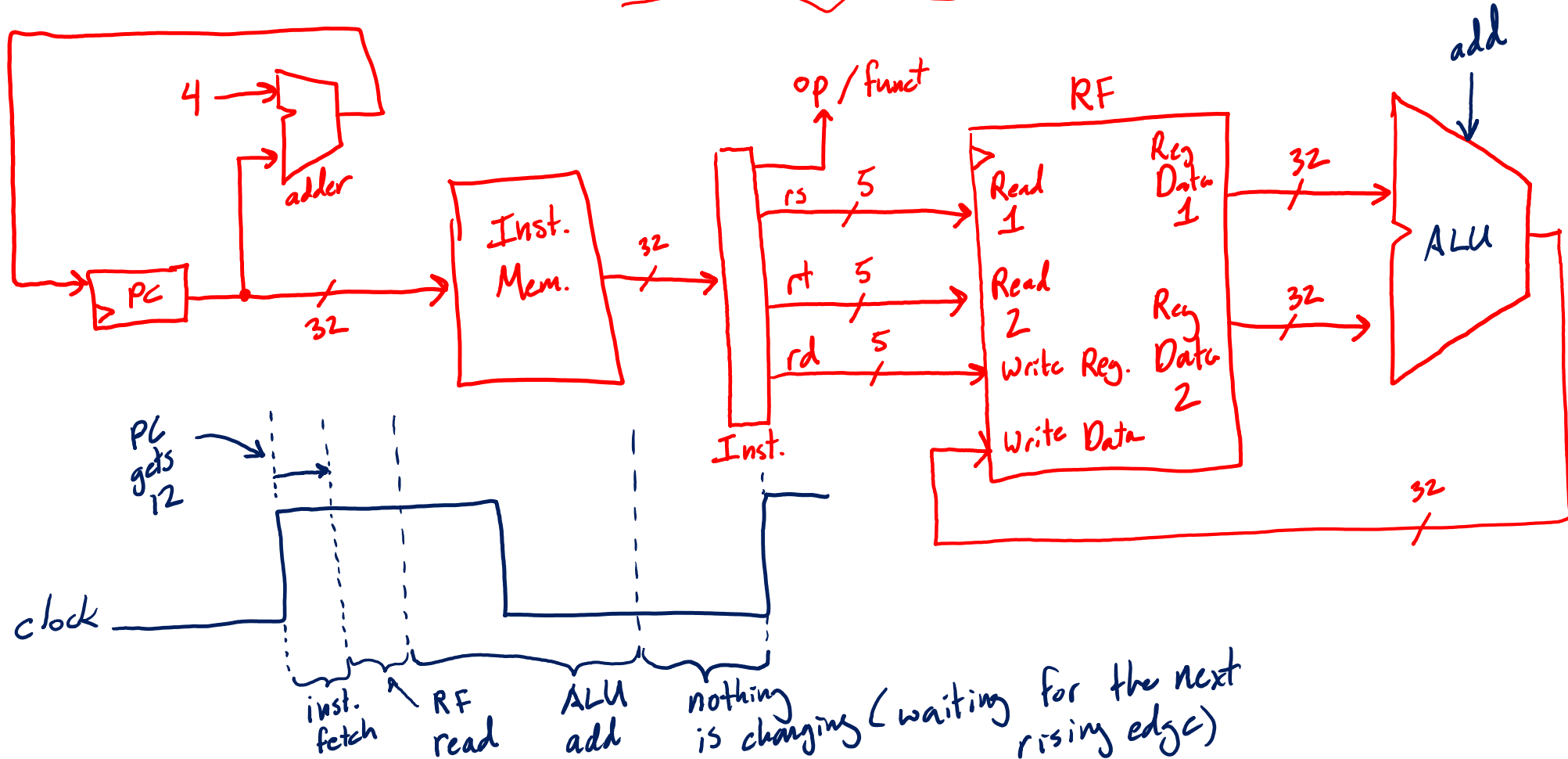


"Add" Instruction

dest.

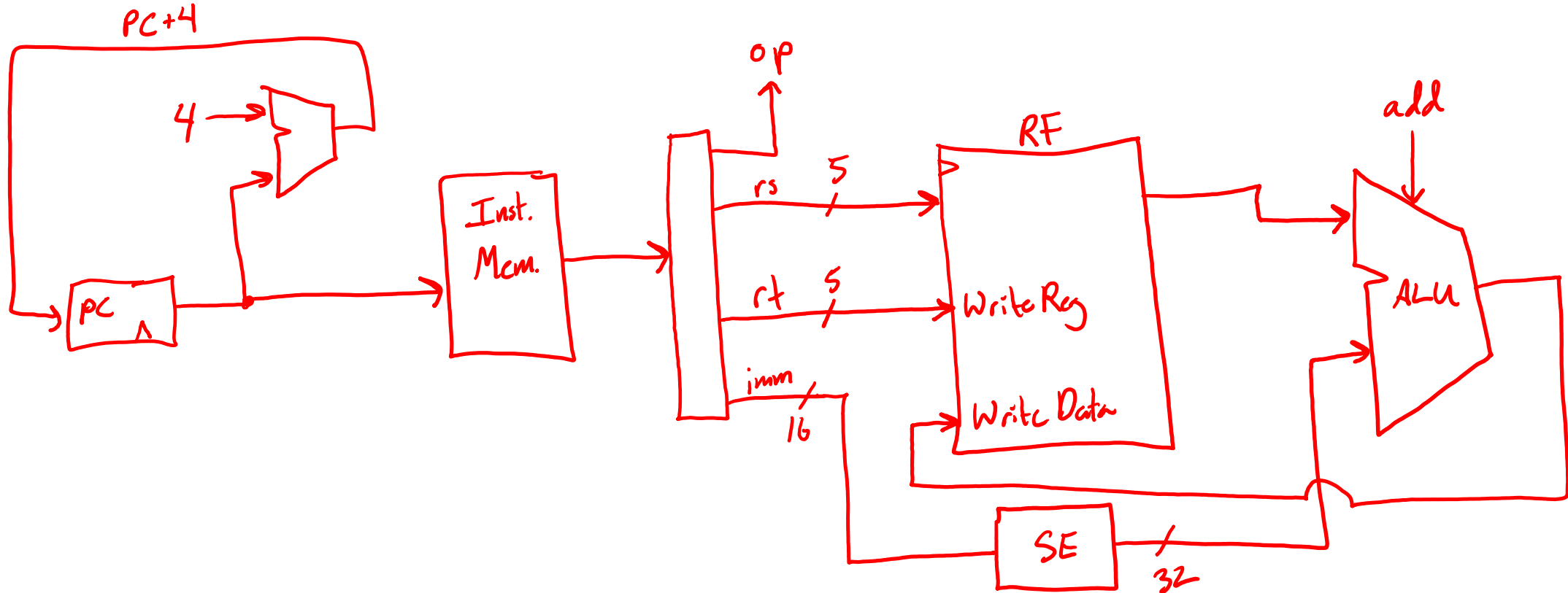
make lab2

Operation	rs	rt	rd	shamt	funct	# meaning
12 add	3	5	2	0	32	# \$2 <- \$3 + \$5



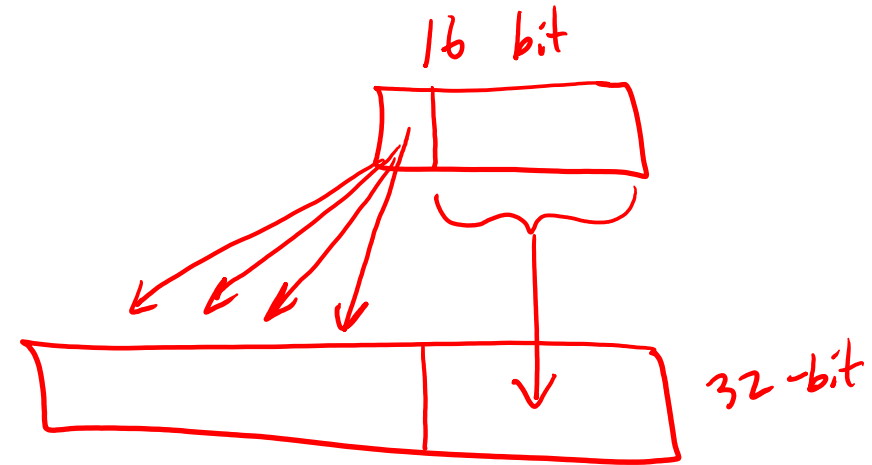
"Addi" Instruction

Operation	rs	rt	imm	# meaning
addi \$5,\$3,6	3	5	6	# \$5 <- \$3 + 6



Sign Extension

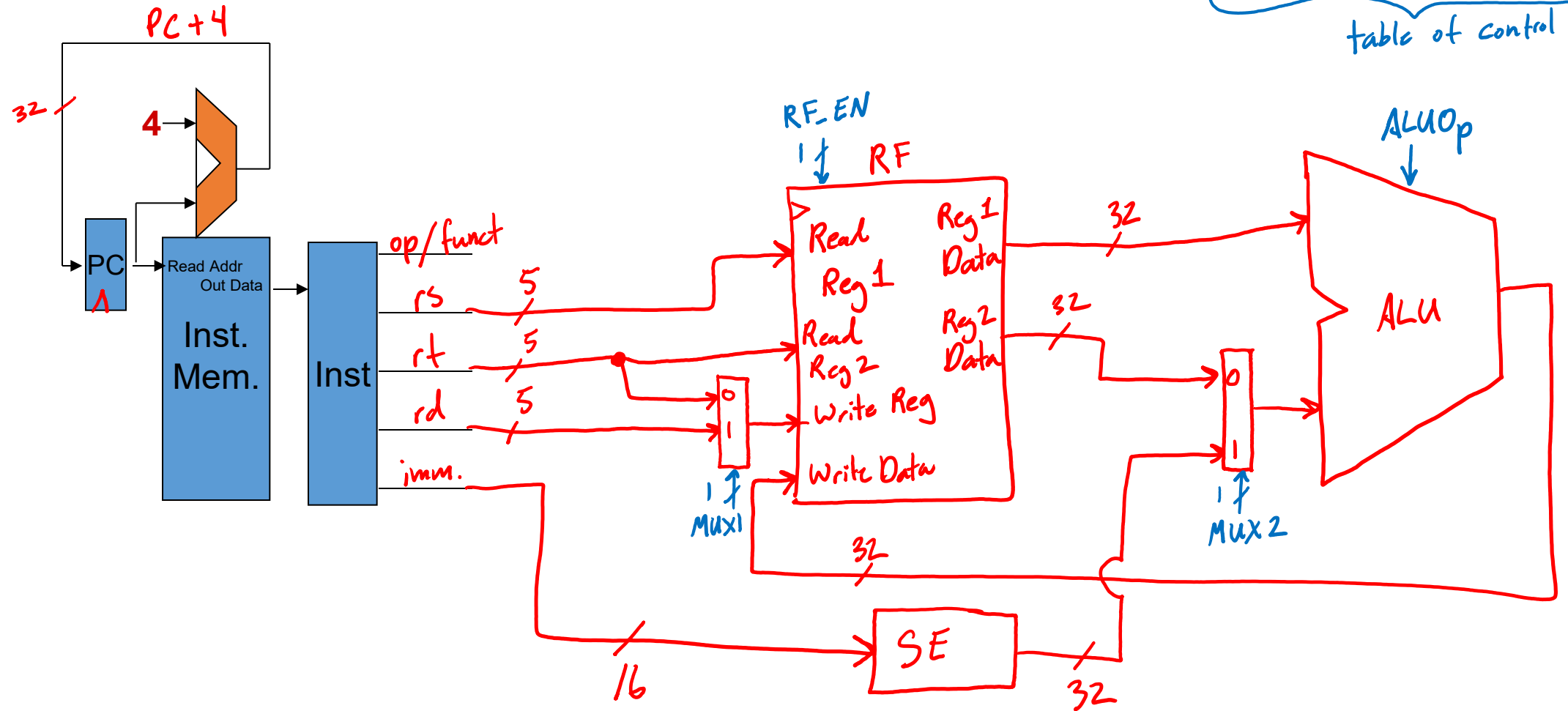
— copy the sign bit



Putting Add & Addi Together

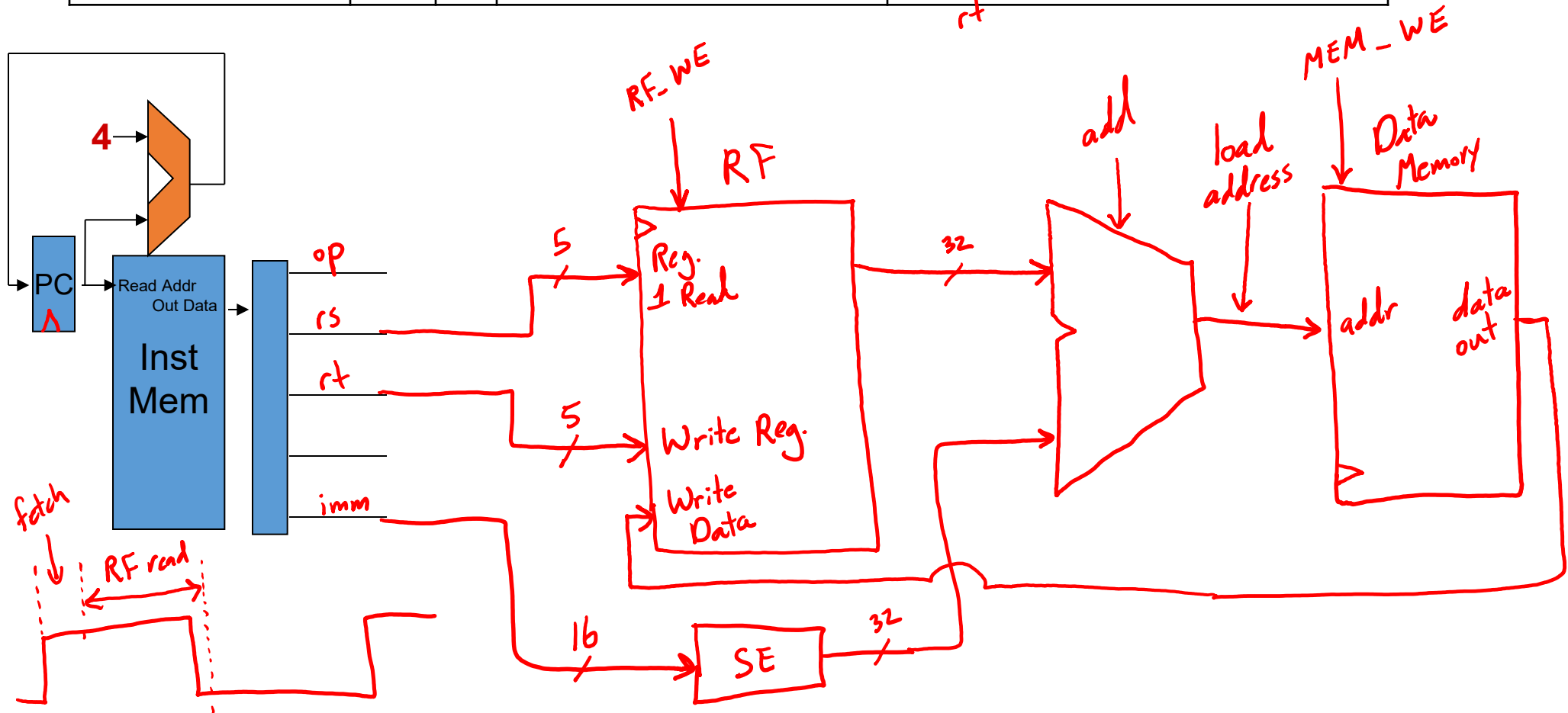
	ALUOp	MUX1	MUX2	RE_EN
add	add	1	0	1
addi	add	0	1	1

table of control signals



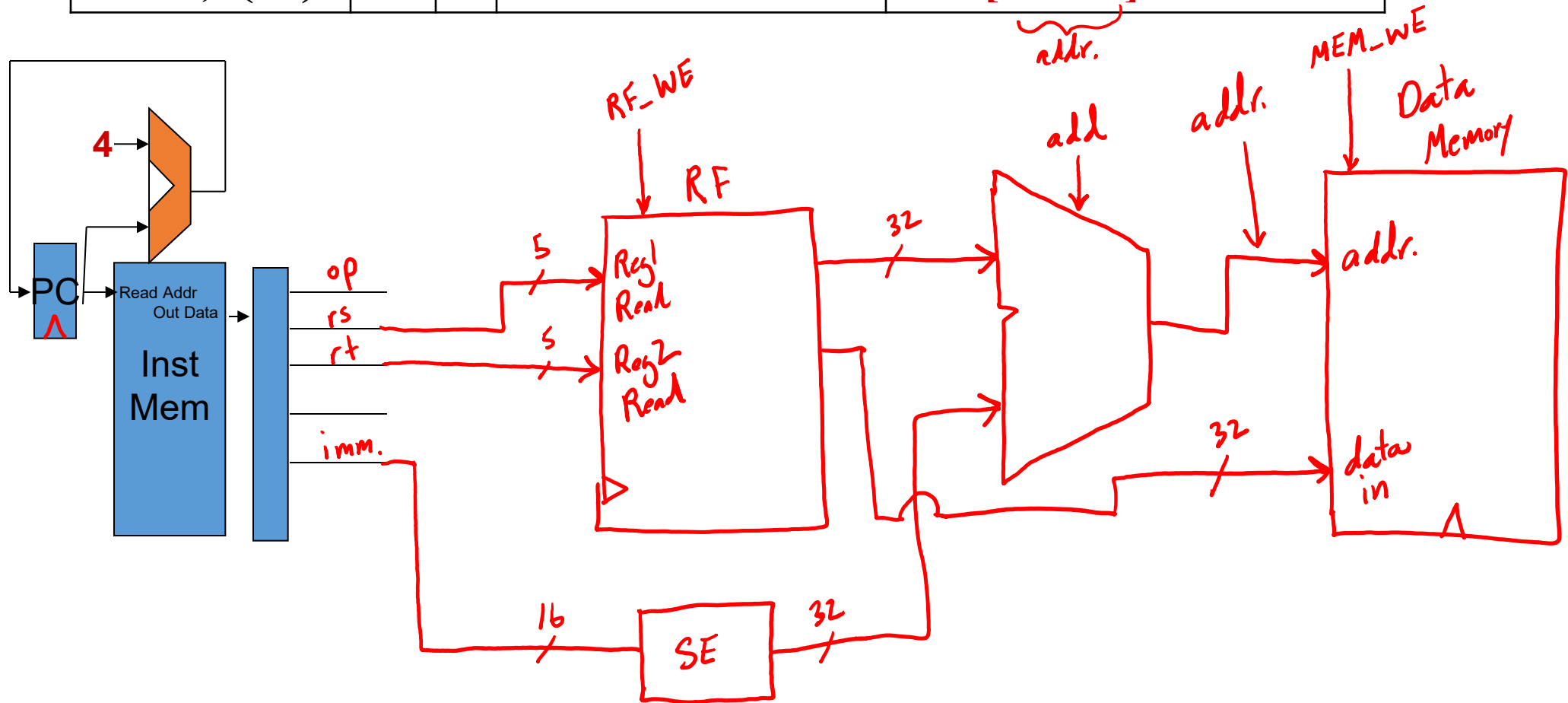
Load Operation

Operation	rs	rt	imm	# meaning
lw \$5,8(\$3)	3	5	8	# $\$5 \leftarrow M[\$3 + 8]$



Store Operation

Operation	rs	rt	imm	# meaning
sw \$5,8(\$3)	3	5	8	# $M[\overset{rs}{\$3} + \overset{imm.}{8}] \leftarrow \overset{rt}{\$5}$



Machine Speed

- What determines the clock rate of a machine?
the time to run the slowest inst.
- Making something bigger makes it: (ex. RF)
 1. code is simpler (nicer software)
 2. may make the clock cycle longer